Chapter-2

Understanding Cloud Computing



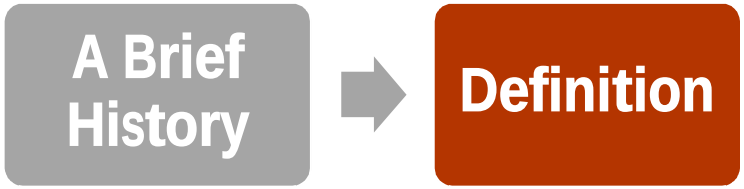
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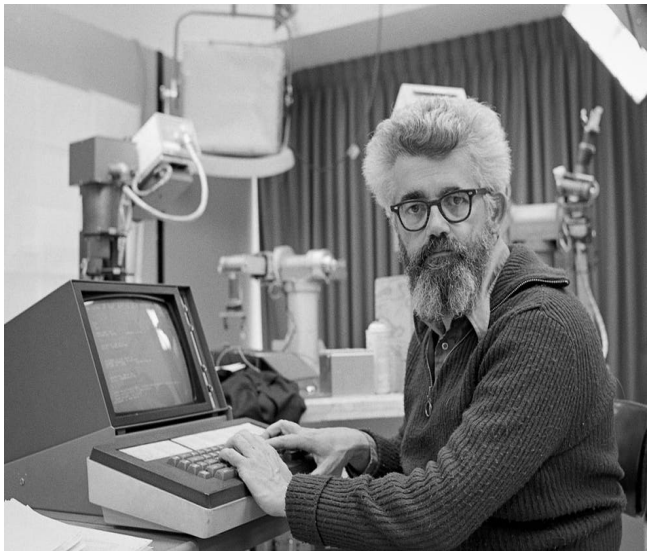
Outlines

- Origins and Influences
- Basic Concepts and Terminology
- Goals and Benefits
- Risks and Challenges

Origins and Influences



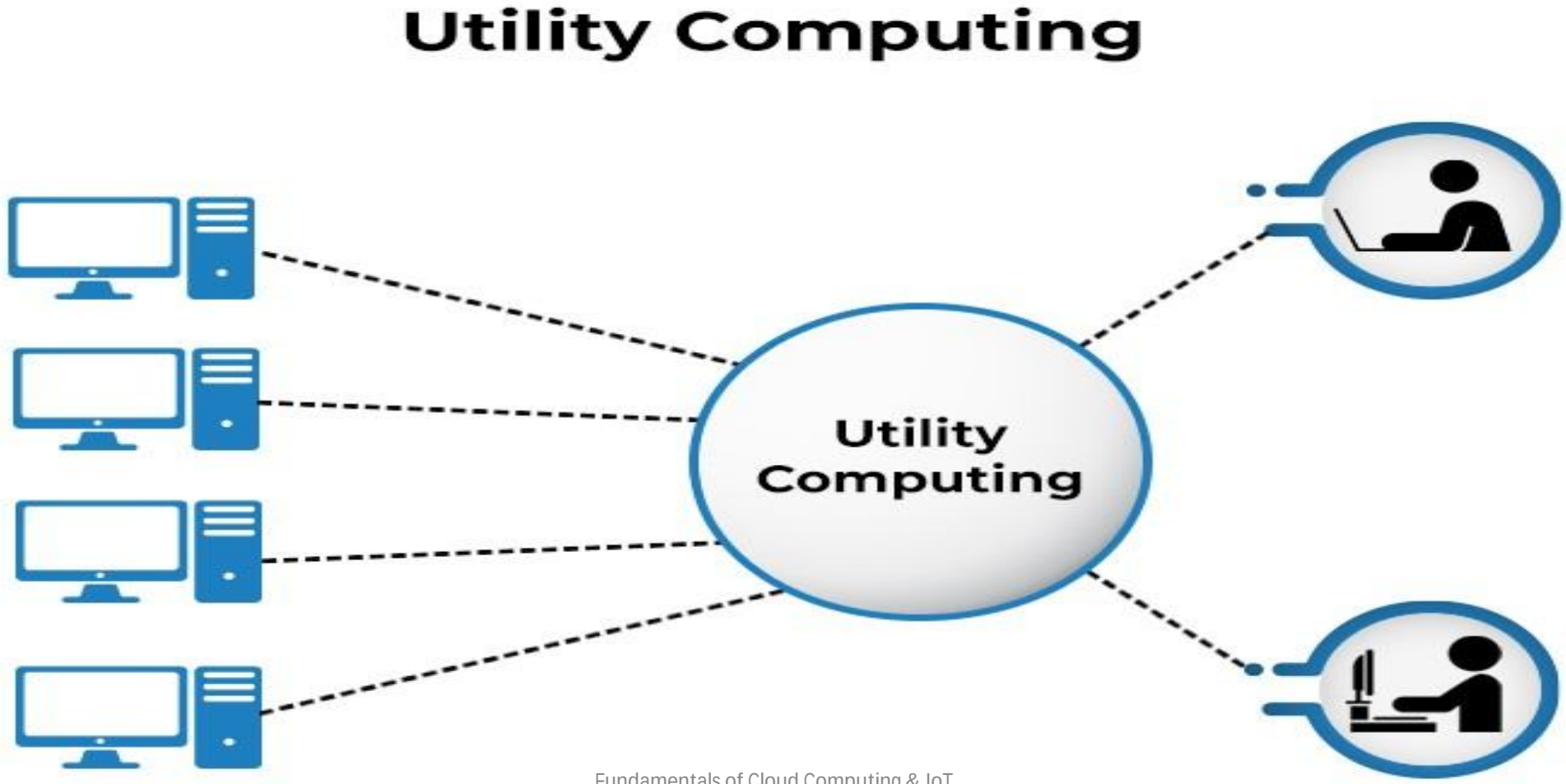
A Brief History



- The idea of computing in a “cloud” traces back to the origins of **utility computing**, a concept that computer scientist **John McCarthy** publicly proposed in 1961:

“If computers of the kind I have advocated become the computers of the future, then computing may someday be organized as a public utility just as the telephone system is a public utility. … The computer utility could become the basis of a new and important industry.”

A Brief History



A Brief History

- In 1969, Leonard Kleinrock, a chief scientist of the Advanced Research Projects Agency Network or ARPANET project that seeded the Internet, stated:

"As of now, computer networks are still in their infancy, but as they grow up and become sophisticated, we will probably see the spread of 'computer utilities' …"



A Brief History

- The general public has been leveraging forms of Internet-based computer utilities since the mid-1990s through various incarnations of







  - ❑search engines (Yahoo!, Google),
  - ❑e-mail services (Hotmail, Gmail),
  - ❑open publishing platforms (MySpace, Facebook, YouTube), and
  - ❑**other types of social media** (Twitter, LinkedIn).
- Though consumer-centric, these services popularized and validated core concepts that form the basis of modern-day cloud computing.

A Brief History

- In the late 1990s, **Salesforce.com** pioneered the notion of bringing remotely provisioned services into the enterprise. i.e. ***CRM***
- In 2002, Amazon.com launched the **Amazon Web Services (AWS)** platform, a suite of enterprise-oriented services that provide remotely provisioned ***storage, computing resources, and business***

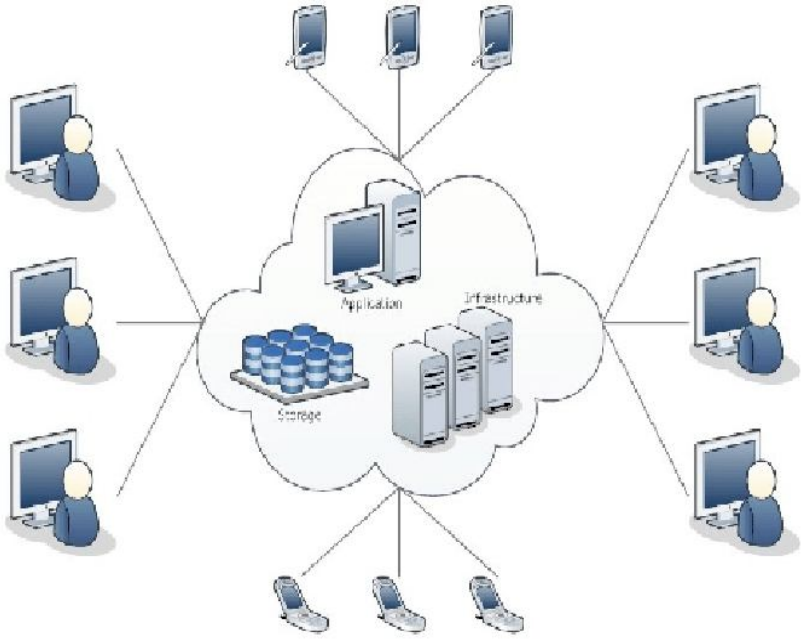


A Brief History

- A slightly different evocation of the term "**Network Cloud**" or "**Cloud**" was introduced in the early 1990s throughout the networking industry.
- It referred to an abstraction layer derived in the delivery methods of data across heterogeneous public and semi-public networks that were primarily **packet-switched**, although cellular networks used the "Cloud" term as well.

A Brief History

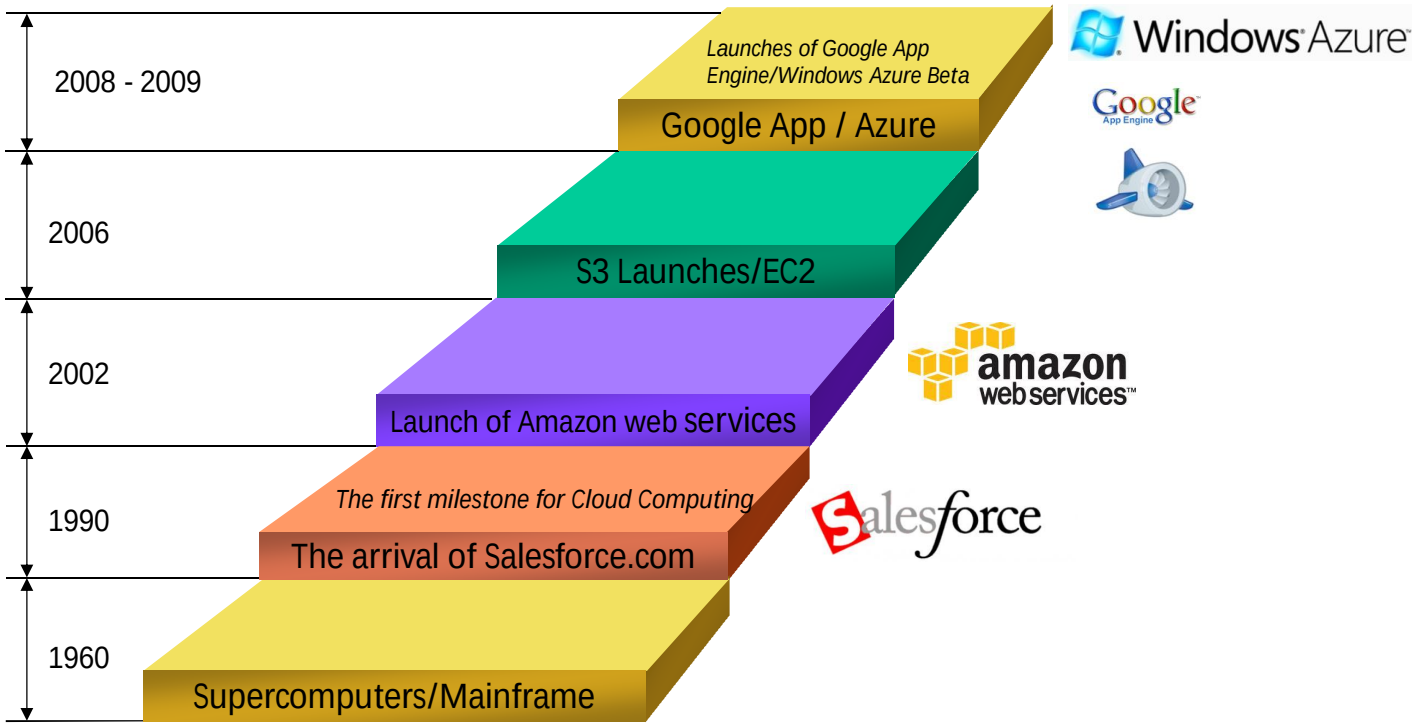
- The networking method at this point supported the transmission of data from one end-point (local network) to the “Cloud” (wide area network) and then further decomposed to another intended end-point.
- This is relevant, as the networking industry still references the use of this term, and is considered an early adopter of the concepts that underlie utility computing.



A Brief History

- It wasn't until 2006 that the term "**cloud computing**" emerged in the commercial arena.
- It was during this time that Amazon launched its Elastic Compute Cloud (EC2) services that enabled organizations to "lease" computing capacity and processing power to run their enterprise applications.
- Google Apps also began providing browser-based enterprise applications in the same year, and three years later, the Google App Engine became another historic milestone.

A Brief History: Summary



Definition Gartner

- A **Gartner report** listing cloud computing at the top of its strategic technology areas further reaffirmed its prominence as an industry trend by announcing its formal definition as:

"...a style of computing in which scalable and elastic IT-enabled capabilities are delivered as a service to external customers using Internet technologies."

- This is a slight revision of Gartner's original definition from 2008, in which "***massively scalable***" was used instead of "***scalable and elastic***."
- This acknowledges the importance of scalability in relation to the ability to scale vertically and not just to enormous proportions.

Definition Forrester

- Forrester Research provided its own definition of cloud computing as:

"...***a standardized IT capability (services, software, or infrastructure) delivered via Internet technologies in a pay-per-use, self-service way.***"

- The definition that received industry-wide acceptance was composed by the ***National Institute of Standards and Technology (NIST).***
- NIST published its original definition back in 2009, followed by a revised version after further review and industry input that was published in September of 2011:

Definition NIST

"Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of five essential characteristics, three service models, and four deployment models."

- Cloud computing is a specialized form of distributed computing that introduces utilization models for remotely provisioning scalable and measured resources.

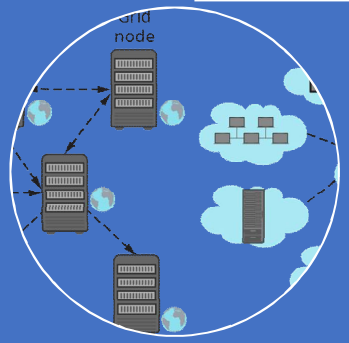
Business Drivers

Capacity Planning

Cost Reduction

Organizational Agility

Technology Innovations



Clustering

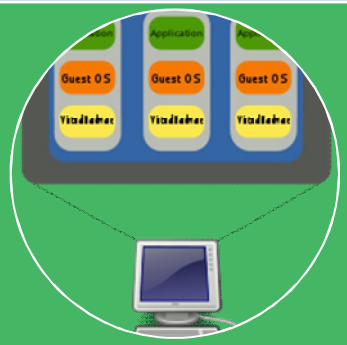
A cluster is a group of independent IT resources that are interconnected and work as a single system.



Grid Computing

A computing grid (or "computational grid") provides a platform in which computing resources are organized

into one or more logical pools.



Virtualization

Virtualization represents a technology platform used for the creation of virtual instances of IT resources



Technology Innovations vs. Enabling Technologies

- **Broadband Networks and Internet Architecture**
- **Data Center Technology**
- **Web Technology**
- **Multitenant Technology**
- **Service Technology**

Basic Concepts and Terminology

Cloud

- **A cloud refers to a distinct IT environment that is designed for the purpose of remotely provisioning scalable and measured IT resources.**
- The term originated as a metaphor for the Internet which is, in essence, a network of networks providing remote access to a set of decentralized IT resources.
- Prior to cloud computing becoming its own formalized IT industry segment, the symbol of a cloud was commonly used to represent the Internet in a variety of specifications and mainstream documentation of Web-based architectures.
- **This same symbol is now used to specifically represent the boundary of a cloud environment.**

Cloud

- **It is important to distinguish the term "cloud" and the cloud symbol from the Internet.**
- As a specific environment used to remotely provision IT resources, a cloud has a finite boundary.
- There are many individual clouds that are accessible via the Internet.
- **Whereas the Internet provides open access to many Web-based IT resources, a cloud is typically privately owned and offers access to IT resources that is metered.**
- Much of the Internet is dedicated to the access of content-based IT resources published via the World Wide Web.

Cloud

- IT resources provided by cloud environments, on the other hand, are dedicated to supplying back-end processing capabilities and user-based access to these capabilities.
- Another key distinction is that it is not necessary for clouds to be Web-based even if they are commonly based on Internet protocols and technologies.
- Protocols refer to standards and methods that allow computers to communicate with each other in a predefined and structured manner.
- A cloud can be based on the use of any protocols that allow for the remote access to its IT resources.

IT Resource

- An IT resource is a physical or virtual IT-related artifact that can be either software based, such as a virtual server or a custom software program, or hardware-based, such as a



Figure Examples of common IT resources and their corresponding symbols.

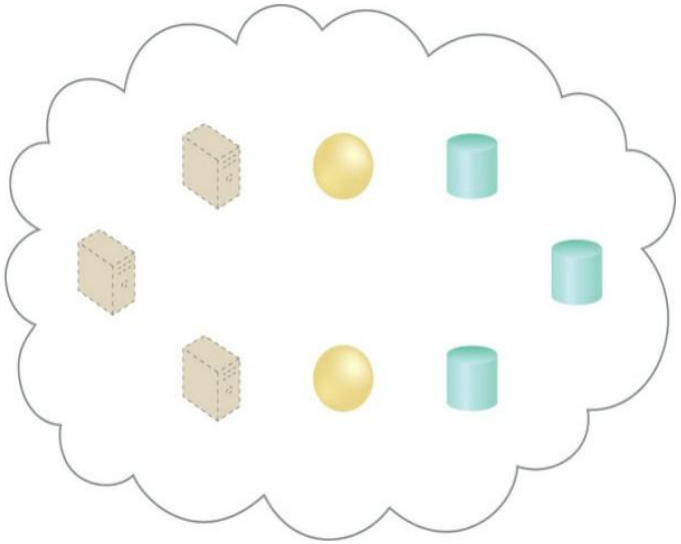


Figure A cloud is hosting eight IT resources: three virtual servers, two cloud services, and three storage devices.

On-Premise

- As a distinct and remotely accessible environment, a cloud represents an option for the deployment of IT resources.
- An IT resource that is hosted in a conventional IT enterprise within an organizational boundary (that does not specifically represent a cloud) is considered to be located on the premises of the IT enterprise, or on-premise for short.
- **In other words, the term "on-premise" is another way of stating "on the premises of a controlled IT environment that is not cloud-based."**
- This term is used to qualify an IT resource as an alternative to "cloud-based."
- An IT resource that is on-premise cannot be cloud-based,and vice-versa.

On-Premise

- Note the following key points:
- An on-premise IT resource can access and interact with a cloud-based IT resource.
- An on-premise IT resource can be moved to a cloud, thereby changing it to a cloud-based IT resource.
- Redundant deployments of an IT resource can exist in both on-premise and cloud based environments.

Cloud Consumers and Cloud Providers

- The party that **provides cloud-based** IT resources is the cloud provider.
- The party that **uses** cloud-based IT resources is the **cloud consumer.**
- These terms represent roles usually assumed by organizations in relation to clouds and corresponding cloud provisioning contracts.

Scaling

from an IT resource perspective, represents the ability of the IT resource to handle increased or decreased usage demands.

The following are types of scaling:

Horizontal Scaling – scaling out and scaling in

Vertical Scaling – scaling up and scaling down

Horizontal Scaling

- The allocating or releasing of IT resources that are of the same type is referred to as horizontal scaling
- The horizontal allocation of resources is referred to as scaling out and the horizontal releasing of resources is referred to as scaling in.
- Horizontal scaling is a common form of scaling within cloud environment

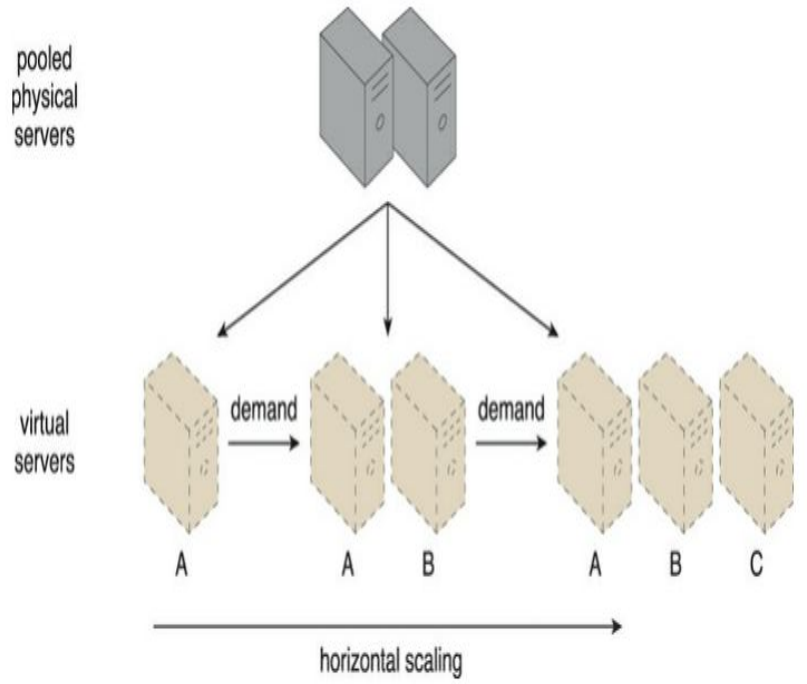
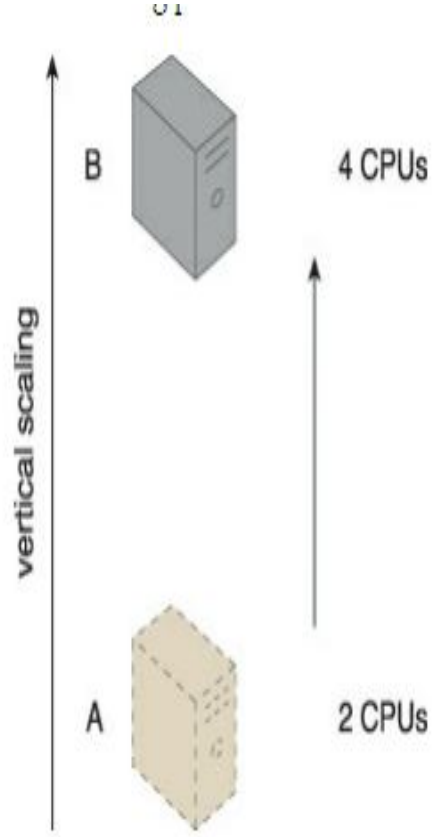


Figure An IT resource (Virtual Server A) is scaled out by adding more of the same IT resources (Virtual Servers B and C).

Vertical Scaling

- When an existing IT resource is replaced by another with higher or lower capacity, vertical scaling is considered to have occurred.
- Specifically, the replacing of an IT resource with another that has a higher capacity is referred to as scaling up and the replacing an IT resource with another that has a lower capacity is considered scaling down.
- Vertical scaling is less common in cloud environments due to the downtime required while the replacement is taking place.



A comparison of horizontal and vertical scaling

Horizontal Scaling	Vertical Scaling
less expensive (through commodity hardware components)	more expensive (specialized servers)
IT resources instantly available	IT resources normally instantly available
resource replication and automated scaling	additional setup is normally needed
additional IT resources needed	no additional IT resources needed
not limited by hardware capacity	limited by maximum hardware capacity

[redacted] A comparison of horizontal and vertical scaling.

Cloud Service

- A cloud service is any IT resource that is made remotely accessible via a cloud.
- Unlike other IT fields that fall under the service technology umbrella such as service- oriented architecture the term “service” within the context of cloud computing is especially broad.
- A cloud service can exist as a simple Web-based software program with a technical interface invoked via the use of a messaging protocol, or as a remote access point for administrative tools or larger environments and other IT resources.

Cloud Service

- A multitude of models for generic types of cloud services have emerged, most of which are labeled with the “as-a-service” suffix
- The driving motivation behind cloud computing is to provide IT resources as services that encapsulate other IT resources, while offering functions for clients to use and leverage remotely.

Cloud Service

- Cloud service usage conditions are typically expressed in a service level agreement (SLA) that is the human readable part of a service contract between a cloud provider and cloud consumer.
- SLA describes QoS features, behaviors, and limitations of a cloud based service or other provisions.
- An SLA provides details of various measurable characteristics related to IT outcomes, such as uptime, security characteristics, and other specific QoS features, including availability, reliability, and performance.

Cloud Service Consumer

- The cloud service consumer is a temporary runtime role assumed by a software program when it accesses a cloud service.
- common types of cloud service consumers can include

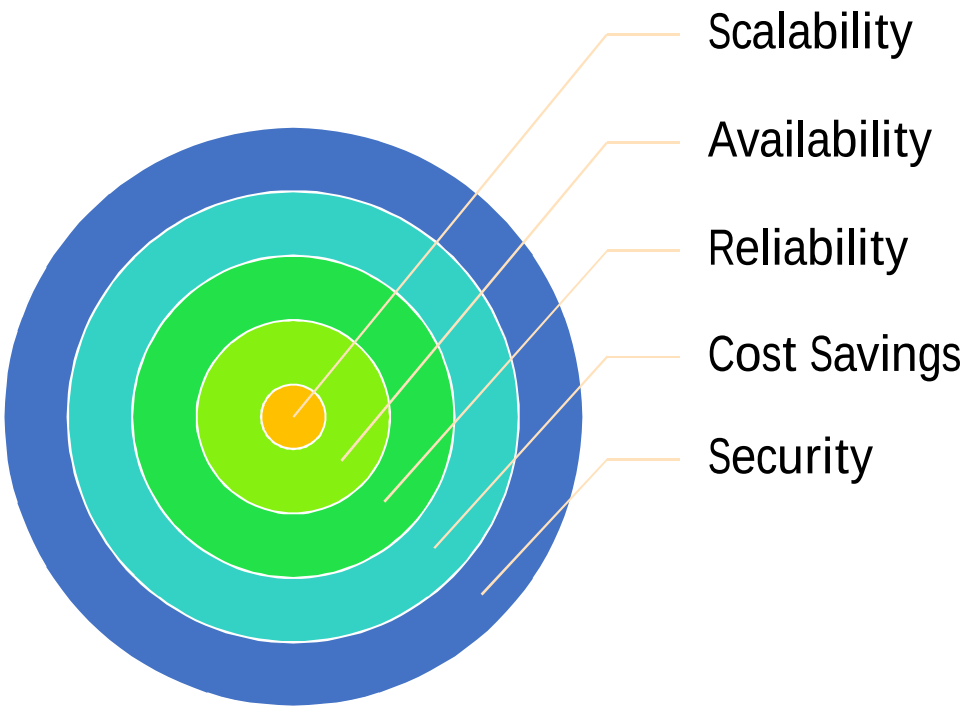


Goals and Benefits

‘Five’ Essential Goals of Cloud Computing

- **Elasticity:** Ability to scale virtual machines **resources up or down**
- **On-demand usage: Ability to add or delete computing power (CPU, memory),** and storage according to demand
- **Pay-per-use:** Pay only for what you use
- **Multitenancy:** Ability to have **multiple customers** access their servers in the data center in an isolated manner
- **Ubiquitous Network access:** ability to be accessed everywhere via any device.

Benefits


Scalability
Availability
Reliability
Cost Savings
Security

Risks and Challenges

Risks and Challenges

Increased Security Vulnerabilities

Cloud Migration

Improper Access Controls and Management

Lack of Expertise

Multi-Regional Compliance and Legal Issues

...etc

Cont…

To be Continued with Lecture-2